

Transformer-Based Person Detection in Paired RGB-T Aerial Images With VTSaR Dataset

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Abstract—Aerial-based person detection poses a significant challenge, yet it is crucial for real-world applications like air-ground linkage search and all-weather intelligent corescuing. However, existing person detection models designed for aerial images heavily rely on numerous labeled instances and exhibit limited tolerance towards complex lighting conditions commonly encountered in search and rescue (SaR) scenarios. This article presents the visible-thermal from SaR scenarios for person detection network (VTSaRNet) to address the challenge of detecting persons situated sparsely in SaR scenes marked by intricate illumination conditions and restricted accessibility. VTSaRNet integrates the instance segmentation for copy-paste mechanism (ISCP) using a Union Transformer Network that functions in both Visible (V) and Thermal (T) bimodalities. Specifically, This study employs synthetic samples obtained through offline Mosaic augmentation by oversampling the local area of bulk images. Then, it utilizes the ISCP module to extract accurate boundaries of personnel instances from complex backgrounds. VTSaRNet cross-integrates the global features and encodes the correlations between two modalities through the multihead attention module. It also adaptively recalibrates the channel responses of partial feature maps for fusion operations with the transformer module in conjunction with anchor-based detectors. Moreover, the adaptation scheme is constructed with multiple strategies to effectively handle various scenarios involving persons, and the entire network is trained end-to-end. Extensive experiments conducted on the Heridal and VTSaR datasets demonstrate the effectiveness of light-weighted VTSaRNet in achieving impressive metrics precision of 98.3%, recall of 96.78%, mAP@0.5 of 98.73%, and mAP@0.5:0.95 of 73.98% under self-built VTSaR dataset, respectively). This performance sets a new benchmark in person detection from aerial imagery.

Index Terms—Aerial-based person detection, bimodality transformer, instance segmentation for copy-paste (ISCP), VTSaR dataset.

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I. INTRODUCTION

PERSON detection in aerial images is a crucial area of computer vision, particularly in constructing search and rescue (SaR) systems. Detecting persons in distress with limited time in a precise and punctual manner is of utmost importance. In recent decades, there has been rapid development and remarkable achievements in aerial-based person detection technology [1], [2], [3], [4], [5], [6], [7], [8], [9]. However, the challenge persists in finding and locating sparsely positioned persons under adverse illumination scenarios for SaR mission. These impairments not only compromise the visual quality subjectively but also undermine the usability and robustness of object detection. Consequently, this poses risks to air-assisted rescue missions and the situational awareness involved in SaR scenarios. For instance, while the authors in [10] and [11] achieve a person detection precision of over 80% on our self-built VTSaR dataset with the best training parameters, it struggles to detect persons concealed in complex SaR scenes, as illustrated in Fig. 1(b).

Recent studies have primarily focused on the development of person detection benchmarks in aerial-based environments. For instance, Pyrrö et al. [14] proposed a novel deep-learning algorithm for automating aerial person detection (APD) during land SaR operations. Zhang et al. [15] introduced an image cropping strategy and attention mechanism based on yolov5 [12] to address small-person detection in the optimized VisDrone2019 dataset [16]. Hollander et al. [17] demonstrated the successful use of patch-based adversarial attacks to camouflage large military assets from automatic detectors in unmanned aerial surveillance. Specker et al. [18] presented a fast and efficient framework tailored to the characteristics of aerial data, specifically focusing on person detection, tracking, reidentification, and their interplay. Vasić and Papić [19] proposed a new approach utilizing sequences of consecutive images as input to a multimodal neural network architecture for person detection. The authors in [20], [21], and [22] emphasize the development of novel technologies in RGB-infrared and multimodal object detection to augment image recognition capabilities of UAVs and to facilitate cross-modal knowledge distillation. While these works contribute to the field of aerial-based person detection, they fail to achieve satisfactory efficacy in complex environments with low light conditions when relying solely on unimodal samples (e.g., RGB or thermal) from drones, particularly in recent datasets, as exemplified in [14] and [23]. Moreover, feature extraction from RGB images proves highly challenging in environments with

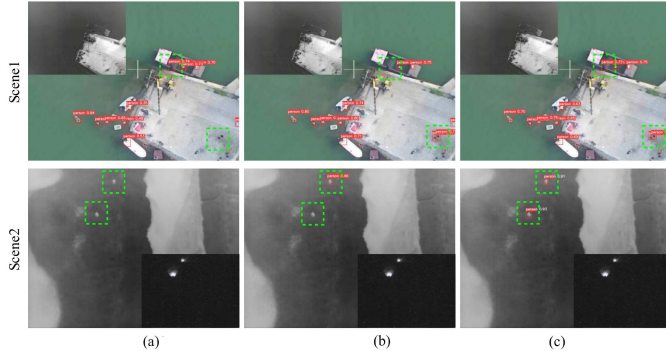


Fig. 1. Visualization of results from detectors of Baseline (YOLOv5 [12]), Cross-Modality Fusion Transformer (CFT) [13], and our proposed method (VTSaRNet). (a) Test is conducted on a single modality, whereas in (b) and (c) test is conducted on fusion dual-modality. Person detection results are marked with red bounding boxes. Subwindow in the left-top corner represents the aligned modality of images, while the green sub-window in the right-top corner magnifies partial results of detection. (a) YOLOv5. (b) CFT. (c) VTSaRNet (Ours).

low light or darkness. However, the use of thermal features as additional data for the detector shows considerable potential. Nevertheless, presently, there exists a scarcity of publicly accessible UAV-based bimodal personal datasets for validating this strategy, warranting further exploration in this field.

Low accuracy in object detection is primarily attributable to the absence of a robust convergence model trained with an adequate number of positive samples, which often results in a high incidence of false detections and consequently inaccurate object localization. Over the years, researchers have introduced various techniques to enhance object detectability. These techniques include downsampling strategies, such as image pyramid construction [24], multiscale feature extraction [25], [26], oriented object detection [27], local region of interest generation [28], and feature extraction for multispectral images [29], [30], [31], [32]. In addition, there have been efforts to improve object feature representation [33] and extraction [34]. Despite these advancements, numerous challenges remain, including issues with object localization accuracy, background noise interference, and increasing computational complexity, all of which contribute to suboptimal detection performance [as depicted in Fig. 1(a)]. Furthermore, low accuracy results from poor-quality images and complex lighting conditions in sparsely populated real-world environments, leading to numerous false alarms and missed objects. Consequently, multisource fusion detection has gained attention among researchers, employing separate feature extractors for RGB and thermal images. Recent fusion methods [35], [36] vary from simple weighted averages to more advanced models, such as multimodal fusion networks and attention mechanisms [10], which leverage the complementary nature of the information [11]. However, challenges related to dual-source image acquisition and inadequate image registration in specialized environments significantly affect the performance of image fusion. This study contributes to the literature by focusing on the precise detection of aerial personnel in all-weather, low-density, and sparse SaR scenarios, enhancing the detector's scene generalization capabilities through the integration

of asymmetric dual-source input images and image synthesis techniques.

This article introduces a person detection framework that enhances detection efficiency in SaR scenarios by integrating resolution-asymmetric RGB-T bimodal features with a transformer-based detector and implementing oversampling augmentation strategies within the VTSaR dataset [illustrated in Fig. 1(c)]. To increase the dataset's size and incorporate more positive samples during training, techniques, such as person oversampling and multiple image stitching, are employed. An aligned dual-source dataset is also created to account for various weather conditions. Moreover, a single-stage detector with a joint codec network is proposed to effectively mitigate the challenges of high false positive and alarm rates in aerial image-based person detection. Validation of the proposed visible-thermal from SaR scenarios for person detection network (VTSaRNet) across three datasets (RGBTDronePerson, Heridal, and VTSaR) yielded encouraging results, exceeding the performance of the latest dual-source APD technologies. The contributions of this study are fourfold as follows.

- 1) Aiming at the problem of complex lighting conditions with sparse instances of person, a joint object-aware density augmentation and dual-modality fusion detector scheme is proposed (VTSaRNet), which generates an extensive array of synthetic samples within SaR scenarios and can significantly enhance the accuracy of person detection by leveraging the supplementary information from dual modalities.
- 2) Generating abundant training samples for specific scenarios by utilizing the ISCP module to segment instances of different scales, and combining them with the Mosaic module to expand persons, which can enhance the performance and effectiveness of person detection in challenging terrains with a sparse density of instances.
- 3) Introducing a bimodality fusion transformer (BFT) module and dual-channel feature attention encoding into the backbone and neck modules of anchor-based detectors, such as YOLOv5, it allows efficient cross-fusion of RGB and thermal image data, thereby handling the difficulties posed by changing lighting conditions.
- 4) Creating a VTSaR dataset for SaR scenarios, which includes varied scenes from suburbs, seas, construction sites, and shores captured from various heights and perspectives.

The rest of this article is organized as follows. Section II provides the motivation based on existing work in relevant areas. Section III outlines the detailed design of our proposed method. Section IV presents the experimental validation of our approach. Finally, Section V summarizes the article and discusses future directions.

II. MOTIVATION

This section first presents the motivation based on issues of existing works. Compared to the category of person in natural scenarios, the SaR environment typically exhibits the characteristics shown in Fig. 2, which is characterized by very small object sizes, sparse availability of person categories, and complex

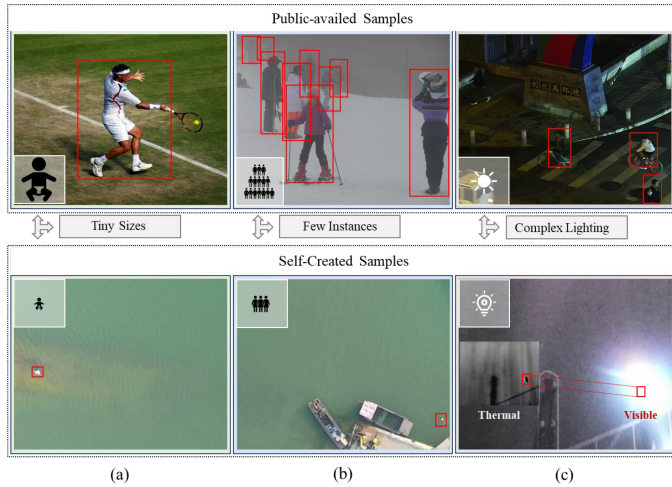


Fig. 2. Comparison of instances captured in natural environments versus aerial scenarios is depicted. The first row introduces samples of persons from the publicly available dataset, while the second row offers instances from the self-generated dataset. For enhanced visual understanding, simulation icons are displayed in the respective lower-left and upper-left corners.

lighting scenes. Therefore, detecting a person in low-altitude images or video sequences using a generic object detector is challenging.

A. RGB-T Dual Aerial Images Alignment

Employing drones equipped with general, economical RGB and infrared sensor cameras for aerial image collection poses several challenges. First, the alignment of visible and infrared images through algorithms becomes difficult due to variations in positioning and field of view when using dual-sensor pods. Second, during the transmission of dual-sensor image signals from mountainous, jungle, or oceanic regions to a terrestrial base, signal strength may diminish, resulting in frame interference, loss, or noise; thus, ensuring stable and reliable transmission of dual-source video streams is critical. Third, RGB images typically possess a higher field of view and resolution, often reaching 1 k, 2 k, or higher, while traditional thermal sensors exhibit limited fields of view and image resolutions, commonly restricted to 480 or 720 p. As a result, the alignment of asymmetric multisource images becomes increasingly complex.

B. Sparse Person Samples for SaR Scenarios

Large-scale aerial SaR imagery often lacks or has rare occurrences of persons. As depicted in Fig. 2(b), victims of drowning in vast seas, missing persons in expansive grasslands, or lost persons in dense jungles. Although current deep learning-based high-precision object detectors primarily rely on numerous extensively labeled examples, the quantity and quality of the available data are critical factors. Consequently, there is an urgent need to address the creation of pseudosamples and enhance detector performance when dealing with a limited number of person instances in constrained scenarios.

C. Inconspicuous Person Feature Under Complex Lighting

Detectors used in various applications must possess superior environmental tolerance and object resilience. In challenging lighting conditions, such as bright light, low light, or complete darkness, relying solely on a single-mode image is insufficient to capture intricate feature information about the objects. As depicted in Fig. 2(c), the RGB features of the object are overwhelmed by the illumination signal in bright light environments, while the thermal features remain unaffected. Conversely, in low-light or no-light scenarios, the thermal modality becomes indispensable. Although it is possible to achieve through image translation [37], the feature information tends to be lost. Consequently, there is strong potential in exploring the joint and complementary learning of multimodal features to elevate person detection performance in diverse illumination environments.

III. METHODOLOGY

This section first describes related issues of SaR systems and presents the details of our proposed method.

A. Overall Framework for Aerial-Based Person Detection

This article primarily addresses two significant challenges in person detection within aerial imagery. The first challenge involves achieving accurate person detection in scenarios characterized by sparsity and a wide field of view. The second challenge pertains to the performance degradation of unimodal detection models in varying lighting conditions. To tackle these issues, a detection framework is proposed that utilizes the learning of blocking sparsity and the progressive fusion of person features through a transformer. This framework, depicted in Fig. 3, consists of two primary modules: object density enhancement, which aims to augment the training sample size by consolidating instances in scenarios with sparse or absent objects, and feature fusion with a transformer, employing YOLOv5 [12] within the widely recognized transformer architecture [38] to encode and decode bimodal images, thereby facilitating the fusion of image pixels at the feature level. Finally, detection results are obtained using a nonmaximum suppression method. A comprehensive overview of the framework is provided in the subsequent sections.

B. Object-Aware With Synthetic Data

Synthetic data has become increasingly popular in various computer vision tasks in recent years, primarily due to its cost-effectiveness in generating large volumes of labeled data [39], [40], [41], [42], [43], [44]. However, the concern over the quality of synthetic data arises from its potential inability to accurately capture the distribution and variations present in real-world data. The disparity between the distributions of real and synthetic data can result in diminished performance when models trained on synthetic data are deployed in real-world scenarios. To mitigate this issue, as shown in Fig. 4, the distribution density is primarily enhanced, and the number of persons is increased by two data synthesis strategies in large-scale SaR scenarios.

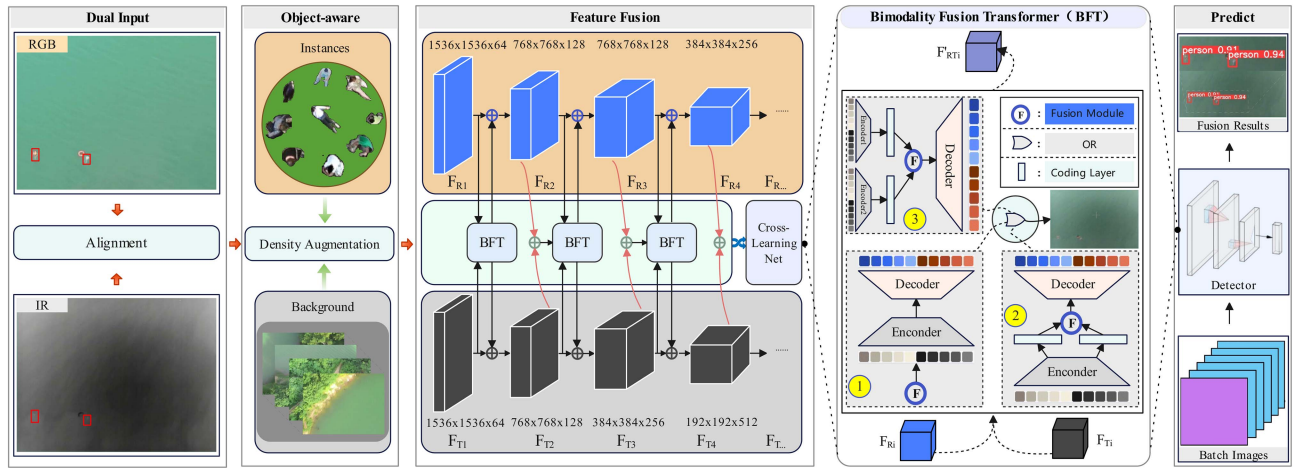


Fig. 3. Overall framework of the proposed method, VTSaNet, is designed for person detection in drone-based scenarios utilizing the self-created VTSaR dataset. This framework primarily comprises two modules: the first module, object density enhancement, incorporates strategies, such as copy–paste with instance segmentation and offline mosaic with block-wise sparse sampling to generate a substantial number of synthetic samples. The second module, feature fusion with the BFT component, integrates cross-modality complementary data from dual aerial image pairs, thereby enhancing the perceptual capabilities of the detection algorithms. All modules are implemented within the unified YOLOv5 framework, with detailed descriptions provided in subsequent sections. In addition, it is important to note that both the input images and the evaluation results are derived from the VTSaRNet, which operates based on the VTSaR dataset.

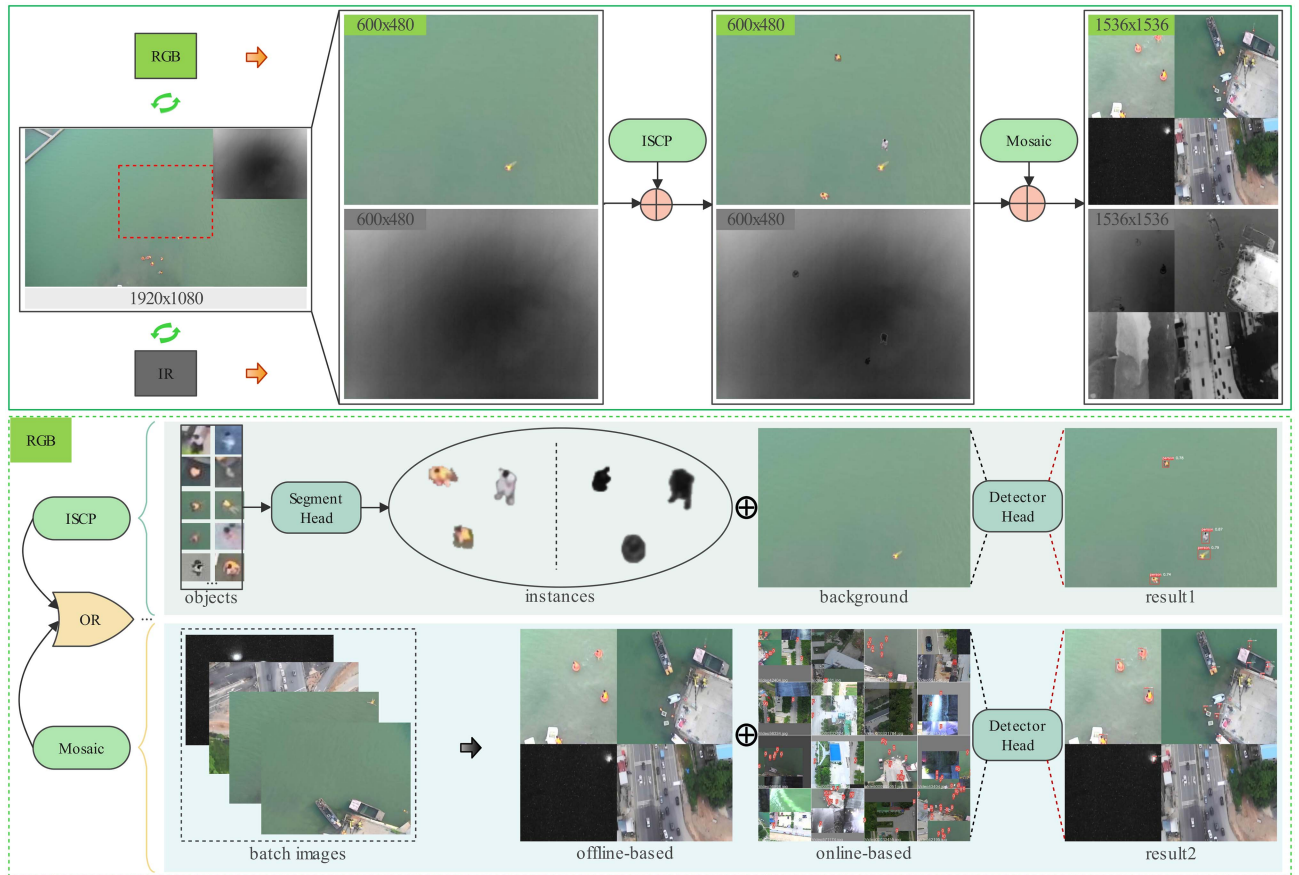


Fig. 4. Illustration of the idea for object implantation with ISCP module and fine-grained detection by Mosaic augmentation. The green solid box depicts the method's overall flow, while the green dashed box depicts the method's execution details in unimodal (RGB).

1) *Copy-Paste With Instance Segmentation*: A copy-paste strategy utilizing the instance segmentation technique is proposed to efficiently expand objects of images [45], [46]. Our approach involves detecting and segmenting the instances of objects of interest in the source image and seamlessly integrating them into the target image while removing the local scene context. As shown in Fig. 4, the ISCP module initially identifies 107 representative crop blocks from a vast patch library of 29 502 instances and segmentation labels for the persons within these crop blocks are manually annotated. The YOLOv5 segmentation head is then employed to train a more robust instance library segmentation model. Subsequently, the trained segmentation model predicts samples from all instance libraries, extracting person masks while removing background information. Moreover, the copy-paste method is used to randomly select a specific number of masks to paste on the existing backgrounds, creating synthetic samples.

2) *Offline Mosaic With Block-Wise Sparse Sampling*: Mosaic is a novel technique proposed in YOLOv4 [47]. The primary concept is to merge four images into one image as training samples and adjust the object bounding boxes' coordinates accordingly during the merging process. Namely, each image undergoes random cropping to select a specific area from the original image (defined as A). In addition, A is resized, along with its coordinate bounding box, to match the output image's dimensions, resulting in B. This ensures a consistent coordinate system size for subsequent merging. Lastly, B is subject to random cropping to create a designated-sized area C, which is then pasted into the corresponding location of the output image. The above description results are shown in Fig. 4 online-based Mosaic module. However, the resulting coordinates after the Mosaic merging are suboptimal. For instance, when assigning positive and negative samples, any intersection over union (IoU) value below a specific threshold between the edge-cropped prediction box and ground truth (GT) is typically considered a negative sample. Furthermore, the randomly selected crop locations in Mosaic may not include positive samples and, hence, do not effectively augment the number of original samples. To address the aforementioned issues, a comprehensive offline-based and online-based Mosaic augmentation strategy is presented, illustrated in Fig. 4. In this approach, the initial step involves merging the original image (640×640), which contains the persons, into a higher resolution intermediate image (1536×1536) through offline-based Mosaic. The intent is to preserve the objects from the original image to the maximum extent possible. Subsequently, batches of small blocks are randomly cropped from the offline-based images to facilitate scale dithering and stitching. Ultimately, the combined outcomes of both approaches are employed to expand objects by feeding them into the detector network.

C. RGB-T Person Detection With Transformer

Integrating cross-modality fused complementary information from pairs of dual aerial images can enhance the perception capability of detection algorithms. In comparison to previous

methods [48], [49], [50], [51], it is believed that distinct features should be processed separately. Modality-specific features should be preserved and enhanced, while the modality-shared features should be selectively extracted from both the RGB and thermal modalities.

1) *Feature-Level Fusion*: In contrast to the initial iteration of YOLOv5, this study presents a novel approach that incorporates a two-stream convolutional neural network backbone. This backbone is illustrated by the light-green and gray modules in Fig. 5, and it is designed to extract feature maps from both the RGB and thermal images. The integration of the BFT module occurs in the fusion stage, when it combines mid-level multi-spectral feature maps derived from both the color and thermal modalities. In the context of fusion utilizing BFT modules, the fusion process occurs at three distinct stages, facilitating the integration of fused characteristics that comprise both multi-scale semantic information and exact spatial details. In addition, the BFT module exhibits versatility and compatibility with many types of two-stream CNN-based detectors when employed as a fusion module.

2) *Multihead Attention*: One shared component in the encoder and decoder structures is the multihead attention [52], depicted on the left side of Fig. 6, comprising parallel self-attention mechanisms. An encoder block contains a single multihead attention, while a decoder block incorporates two multihead attention mechanisms, with one employing a Mask. On the right side of Fig. 6, we see the scale dot product attention, constituting the bedrock of the transformer structure.

Self-attention constitutes three vectors—Q, K, and V, calculated on an identical input, such as a sequential word. Precisely, these vectors, Q, K, and V, are derived via a linear transformation, like deploying a fully connected layer, on the initial input word's embedding. These three vectors typically share dimensions, based on the stipulations made during model design. The calculation of self-attention deploys Q, K, and V to compute attention scores that mirror the relationship between the current position and other positions. Such scores are derived by dotting Q and K, dividing these scores by 8, and normalizing via softmax. This procedure yields weights for each of the positions. Subsequently, these weights serve to compute the weighted sum of V, forming the output for the present position.

3) *Detailed BFT Computation*: To illustrate the efficacy of our proposed BFT fusion module, YOLOv5's feature extraction network is extended and redesigned as the backbone consisting of two streams to enable modal fusion and interaction. The sentences $\mathbf{L}_V \in \mathbb{R}^{HW \times C}$ and $\mathbf{L}_I \in \mathbb{R}^{HW \times C}$ are obtained by flattening each feature map and rearranging the matrices. This process is applied to the intermediate RGB convolution feature maps $\mathbf{F}_V \in \mathbb{R}^{C \times H \times W}$ and the thermal convolution feature maps $\mathbf{F}_I \in \mathbb{R}^{C \times H \times W}$. Subsequently, the sentences originating from each modality are combined and enhanced with a positional embedding that may be adjusted during training. This embedding is a trainable parameter with dimensions of $2HW \times C$, ultimately yielding the input sentences $\mathbf{L} \in \mathbb{R}^{2HW \times C}$ for the transformer model. The BFT module is depicted in Fig. 5. The incorporation of a positional embedding enhances the model's

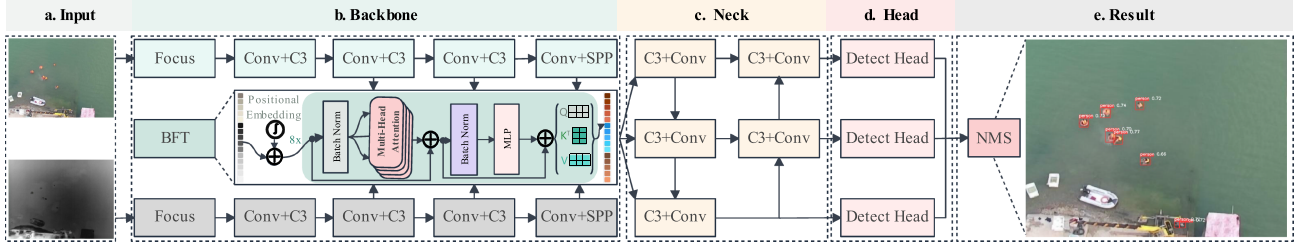


Fig. 5. Architecture of YOLOv5 with the BFT module consists of five distinct components. The primary component is a dual-stream network integrated into the Backbone module, responsible for extracting features from both RGB and thermal modalities. The Neck module employs a divide-and-conquer strategy to achieve feature fusion. Utilizing a feature pyramid inspired by the PANet framework, this approach aggregates features across multiple scales and depths. The BFT module combines mid-level multispectral feature maps from both color and thermal modalities at three distinct stages, facilitating the integration of fused characteristics that encompass both multiscale semantic information and precise spatial details. This process enhances both spatial and semantic information. Subsequently, the detection head performs object detection, followed by postprocessing procedures that include NMS.

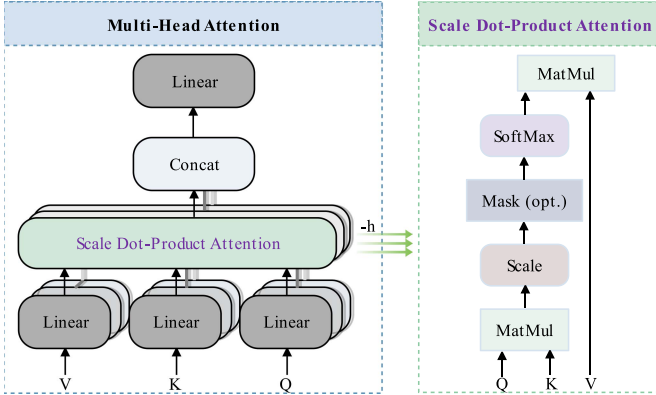


Fig. 6. Concept of multihead attention involves the partitioning of the model into many heads, so establishing distinct subspaces that enable the model to selectively attend to various facets of information. The multihead attention mechanism executes the scaled dot-product attention procedure H times and subsequently combines the resulting outputs.

capacity to differentiate spatial information across tokens during the training process. Next, the input sequence $\mathbf{L}$ is subjected to a projection by means of three weight matrices in order to generate a set of queries, keys, and values ($\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$)

$$\mathbf{Q} = \mathbf{L}\mathbf{W}^Q \quad (1)$$

$$\mathbf{K} = \mathbf{L}\mathbf{W}^K \quad (2)$$

$$\mathbf{V} = \mathbf{L}\mathbf{W}^V. \quad (3)$$

The weight matrices $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$ are defined, where $\mathbf{W}^Q$ is a matrix of dimensions $C \times D_Q$, $\mathbf{W}^K$ is a matrix of dimensions $C \times D_K$, and $\mathbf{W}^V$ is a matrix of dimensions $C \times D_V$. Furthermore, within the self-attention layer (see Fig. 6), the computation of attention weights involves the scaled dot product of matrices $\mathbf{Q}$ and $\mathbf{K}$. These weights are then multiplied by the values to yield the enhanced output. The symbol $\mathbf{X}$ is used to represent a mathematical object

$$\mathbf{X} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_K}} \right) \mathbf{V}. \quad (4)$$

The scaling factor $\frac{1}{\sqrt{D_K}}$ is utilized to prevent the softmax function from encountering extremely small gradients when the dot product magnitudes become large. Furthermore, the incorporation of the multihead attention mechanism facilitates the integration of numerous intricate connections across diverse subspaces and positions of representation

$$\mathbf{X}' = \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\mathbf{Z}_1, \dots, \mathbf{Z}_h) \mathbf{W}^O \quad (5)$$

$$\mathbf{X}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V). \quad (6)$$

The symbol h as a subscript is used to denote the number of heads. The matrix $\mathbf{W}^O \in \mathbb{R}^{h \cdot C \times C}$ represents the projected matrix, which is created by concatenating the matrices $\mathbf{X}_1, \dots, \mathbf{X}_h$. The transformer model utilizes a fully connected feed-forward network with two layers. This network incorporates a rectified linear unit (ReLU) activation function to generate the output sequences $\mathbf{O}$. The output sequences retain the same shape as the input sequences $\mathbf{L}$

$$\begin{aligned} \mathbf{O} &= \text{MLP}(\mathbf{X}'') + \mathbf{X}'' \\ &= \text{FC}_2(\text{RELU}(\text{FC}_1(\mathbf{X}''))) + \mathbf{X}''. \end{aligned} \quad (7)$$

Attention please, $\mathbf{X}'' = \mathbf{X}' + \mathbf{L}$ in this case. The last step is to turn the output lines $\mathbf{O}$ into recalibration results $\mathbf{F}'_V$ and $\mathbf{F}'_I$. Then, using the opposite process of the first step, these results are added to the original modality branch as extra information.

IV. EXPERIMENTS

This section critically evaluates the performance of several notable object detectors, along with confirming the efficiency of the VTSarNet. The initial part of this research details the experimental protocol, which includes the datasets used, parameter configurations, and selected evaluation criteria. Following this, an exhaustive benchmark featuring a variety of detectors is provided. The concluding part of this section serves to endorse the effectiveness of the VTSarNet through an extensive array of comparative experiments and discussionw studies.

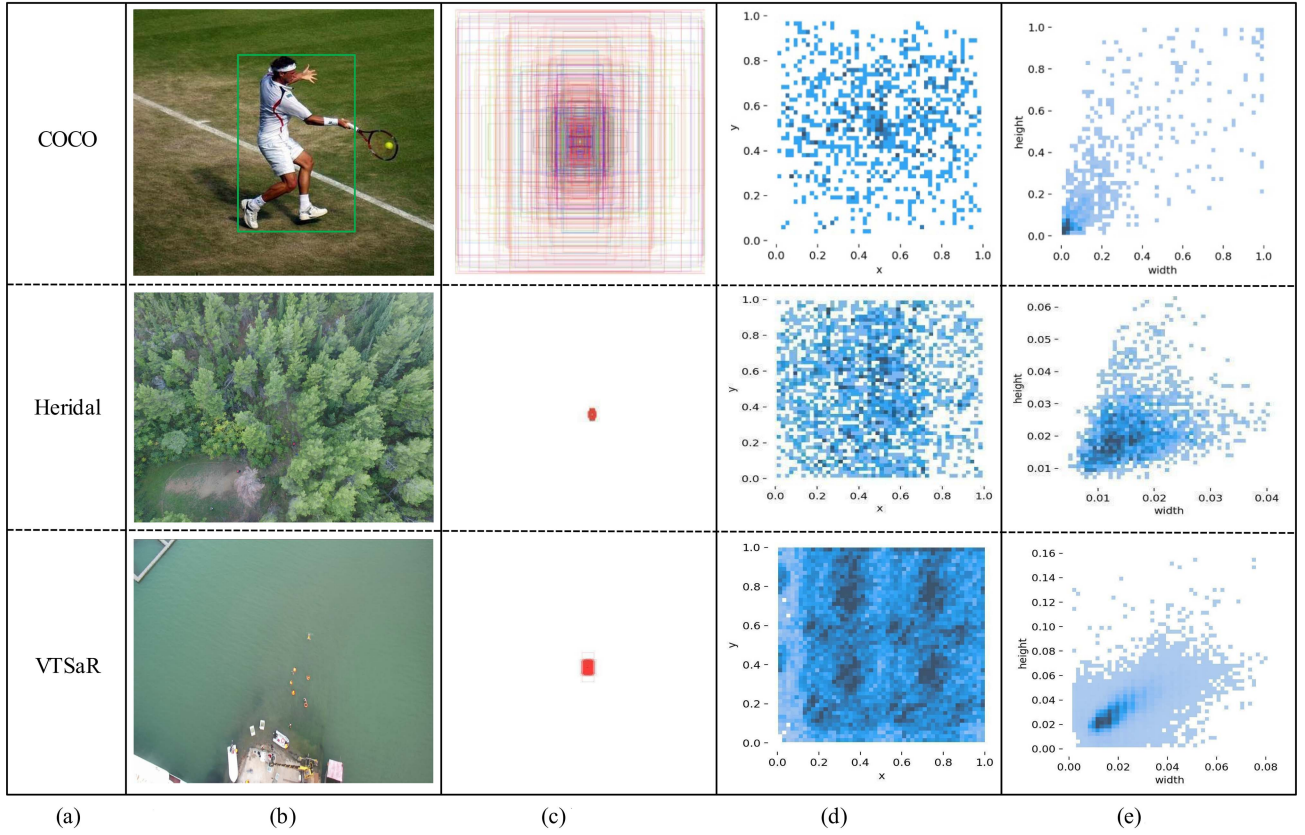


Fig. 7. Comparison of instances across multiple datasets, namely: (a) MS-COCO, Heridal, and VTSaR datasets. (b) Analysis of sample distribution across different SaR scenes. (c) Examination of statistical characteristics of the GT pertaining to the person category. (d) Investigation of statistical properties of the coordinates (center point $[x, y]$) associated with the person category. (e) Evaluation of the aspect ratio ($[width, height]$) statistics of the person category.

A. Datasets and Parameters Setting

1) *Datasets*: In our aerial images, due to the long shooting distance and large angle changes, the size of the object in the image might be reduced, and the proportion of pixels occupied by the personnel object can be very small. Such an object can be defined as a “small object.” Typically, an object that is much smaller than the overall image size or occupies only a small number of pixels in the entire image can be considered a small object (following the benchmark established by the MS-COCO dataset, Objects with dimensions less than 32×32 pixels are typically classified as “small objects”). The following provides descriptions of the publicly accessible Heridal dataset and the self-built VTSaR dataset. Fig. 7 displays the following statistical and properties of annotated sample boxes.

- 1) VTSaR dataset¹ offers a diverse collection of scenes (see Fig. 8) depicting urban areas, seas, shores, and jungles. Samples were collected from a variety of locations using our custom-built unmanned helicopter equipped with a dual-camera gimbal system (see Fig. 9). These samples comprise aerial scenarios diverse in geography, time, and environmental conditions. To ensure variety and representativeness, we chose images depicting persons engaging

in a range of activities, such as standing, walking, and working, against an array of backdrop settings. In the dataset annotation stage, preliminary screening processes were utilized to discard images plagued with blurriness or obstructions. Thereafter, persons in the aerial images were identified and their positions were annotated by bounding boxes. VTSaR comprises 17 video clips, 41 269 frames, and 4099 still images, all annotated with single labels for aligned dual-modality images. The details of the dataset are shown in Table I.

- 2) RGBTDronePerson dataset [53] is the first large-scale benchmark for personnel detection and tracking applications of drones in RGBT (visible and infrared) environments. This dataset encompasses images captured from various scenes, lighting conditions, and weather scenarios. Specifically, the RGBTDronePerson detection dataset contains 6125 pairs of RGBT images, totaling 70 880 instances, all of which were collected using a UAV equipped with an RGBT camera. The dataset covers diverse lighting conditions (including day and night), seasonal variations (from winter to summer), as well as different UAV flight altitudes (ranging from 50 to 80 m) and shooting angles (from 45° to 60° , with a few images captured from a vertical perspective).

¹<https://github.com/zxq309/VTSaR>

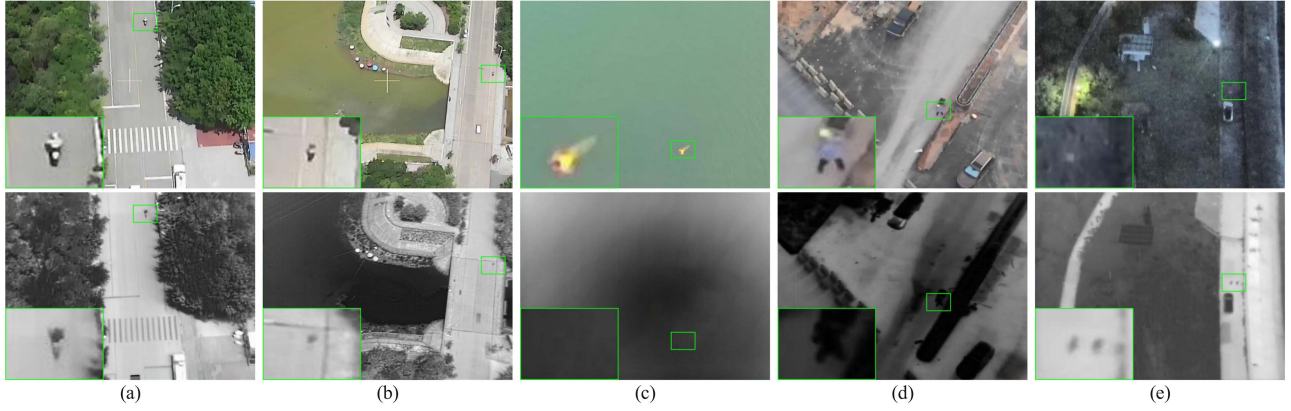


Fig. 8. Visualization of samples (person) under (a)–(e) different environment, where the first row is an exemplar of the RGB mode, while correspondingly, the second row represents an exemplar of the thermal mode.

TABLE I
PARAMETER CONFIGURATION FOR SELF-CREATED VTSAR DATASET IN DRONE-BASED SCENARIOS

Samples	Description	Categories	Size (pix)	Flight Height (m)	Flight Speed (m/s)	Shooting Angle (°)	number of samples			
							Train Set	Val Set	Test Set	Total
Images	Original captured images	person	1920x1080	50, 100, 150, 200, 250, 300	12–15	45, 60, 75	8699	1204	2562	12465
	Cropped sub-images from original images	person	600x480	50, 100, 150, 200, 250, 300	12–15	45, 60, 75	3260	479	1062	4801
	Extended images with Mosaic augmentation	person	1536x1536	50, 100, 150, 200, 250, 300	12–15	45, 60, 75	3260	479	1062	4801
Instances	Original instances from original images	person	12x18, 17x23, 29x24, 31x16 etc.	50, 100, 150, 200, 250, 300	12–15	45, 60, 75	20314	2785	5263	28362
	Extended instances with Copy-Paste	person	Fixed size<(32x32)	50, 100, 150, 200, 250, 300	12–15	45, 60, 75	25590	3492	7525	36607

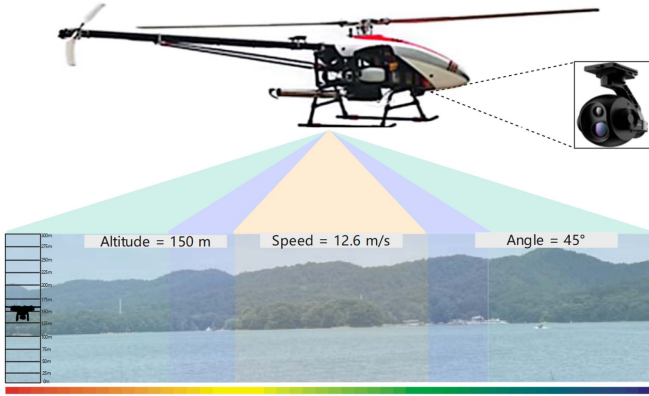


Fig. 9. Examples of acquisition equipment and scenarios by unmanned helicopters.

3) Heridal dataset [14], which is a compilation of 1650 high-resolution images (4000×3000) captured by unmanned helicopters. These images depict persons in nonurban environments, such as mountains, forests, oceans, and deserts.

2) *Image Alignment*: The initially collected dual-source image samples are stored in a picture-in-picture format. First, the same feature points or regions are selected in the two images by marking key points, which serve as reference points for alignment. Subsequently, the necessary geometric transformations—such as translation, rotation, and scaling—are calculated based on the marked key points. The computed transformation is then applied to one of the images to achieve alignment with the

other. Finally, the aligned images undergo visual inspection and evaluation to ensure accuracy and consistency.

3) *Parameters Setting*: The experiments were carried out on a Red Hat 4.8.5-44 operating system, utilizing the following hardware and software specifications: The system configuration includes an Intel Xeon Gold 6134 CPU operating at a frequency of 3.20 GHz, together with two NVIDIA GeForce Tesla V100 GPUs, each equipped with 16 GB of RAM. The backbone utilized in this study is the pretrained ResNet-101 model supplied by ImageNet. The training process involves training all models for a total of 300 epochs. This is achieved by utilizing the stochastic gradient descent optimizer with a momentum value of 0.9 and weight decay set to 0.0001. The initial learning rate was established as 0.01, and a decline was implemented at epoch 10. The hyperparameters utilized in the trainer were tuned through the application of a genetic algorithm [54]. Fig. 10 depicts the optimization procedure involving a comprehensive set of 28 measures. The calculation of the mean average precision (mAP) at validation (mAP@val) was performed for both single-model and single-scale assessment on the aforementioned datasets. In the inference phase, a predetermined score of 0.25 was employed to eliminate background bounding boxes. In addition, an IoU threshold of 0.45 to identify and retain the most confident bounding boxes, limited to a maximum of 3000. Unless explicitly stated, all experiments were performed using the training and inference parameters described before.

4) *Evaluation Metrics*: Object detection evaluation metrics can be categorized into prediction metrics, such as IoU, and classification metrics, such as precision and recall. Precision and recall provide insights into accurate and inaccurate instance

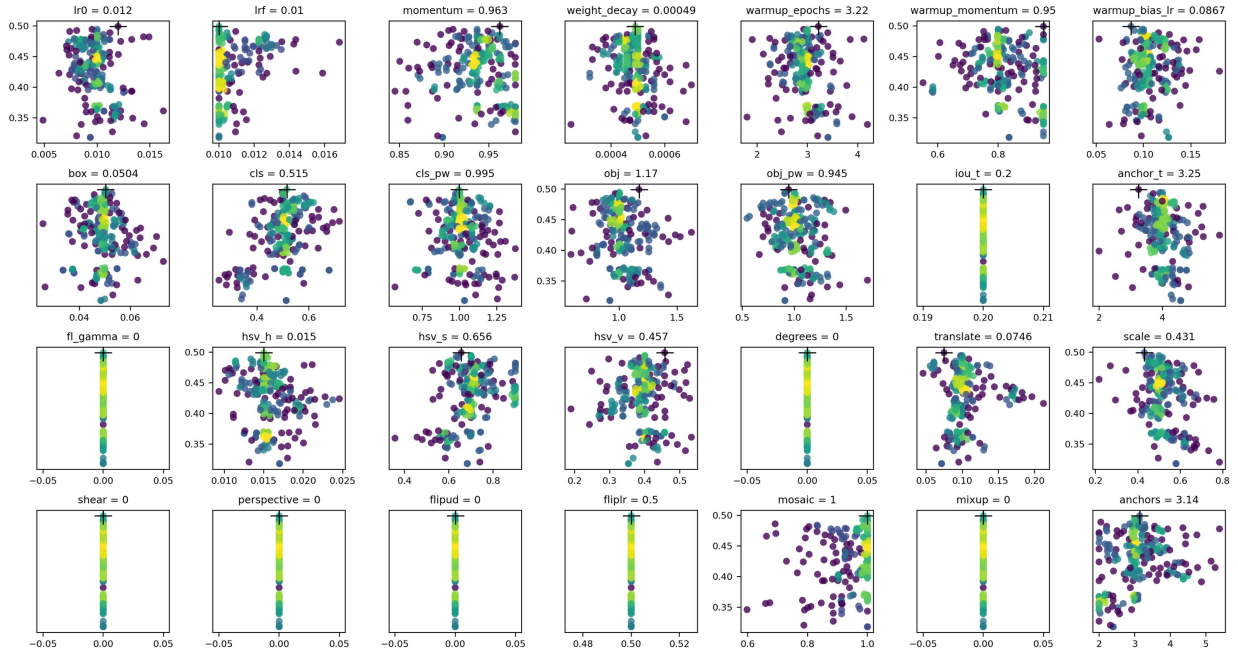


Fig. 10. Visualization results of hyperparameters evolving. One chart for each hyperparameter. The x-axis represents hyperparameter values, whereas the y-axis represents fitness. A higher concentration is indicated by the yellow color. Vertical lines imply that a parameter is unchangeable.

TABLE II
COMPARATIVE RESULTS OF VARIOUS MODULE CONFIGURATIONS AND SOTA METHODS UNDER DIFFERENT PRETRAINED MODELS FOR RGB AND THERMAL MODALITIES ON THE RGBTDRONEPERSON DATASET

OA&Our Methods	InputSize /pix	BackBone	PreTrainedModel	Precision /%	Recall /%	AP ^{val} @0.5 /%	AP ^{val} @0.5:0.95 /%	FPS
RetinaNet	640x512	ResNet50-FPN	retinanet_r50_fpn_1x_coco	57.19	50.14	29.89	20.67	19.4
Faster RCNN	640x512	ResNet50-FPN	faster_rcnn_r50_fpn_1x_coco	56.32	49.52	28.37	19.52	15.9
Faster RCNN	640x512	ResNet50+PANet+FPN	faster_rcnn_r50_panet_fpn_1x_coco	57.47	50.26	29.46	20.83	14.6
Cascade	640x512	ResNet50	cascade_r50_caffe_fpn_gn_head_1x_coco	58.32	51.56	31.55	21.98	11.8
CFT	640x512	CSPDarknet53-PANet	yolov5s	52.36	46.88	22.83	15.86	10.8
QFDet	640x512	ResNet50+FPN	faster_rcnn_r50_fpn_1x_coco	60.11	54.29	42.08	22.47	14.2
YOLOv5 /rgb modality	640x512	DarkNet-53+ Focus	yolov5n	51.27	46.45	20.17	13.55	27.2
YOLOv5 [12] /rgb modality	640x512	DarkNet-53+ Focus	yolov5s	50.04	47.21	21.85	17.42	18.4
VTSaRNet	+ Aug (ISCP)	DarkNet-53+ Focus	yolov5s	58.04	51.77	31.55	22.57	17.3
	+ Aug (Offline-Mosaic)	DarkNet-53+ Focus	yolov5s	58.75	52.63	33.38	26.40	16.8
	+ BFT	DarkNet-53+ Focus	yolov5s	59.84	52.94	36.74	28.79	12.1
	Aug (ISCP+Offline-Mosaic) + BFT	DarkNet-53+ Focus	yolov5s	60.79	53.66	40.42	30.64	10.5

predictions within the overall dataset. In the training stage of most contemporary object detectors, a default IoU of 0.45 is commonly used. To assess the detection performance of VTSaRNet, the mean average precision (mAP) metric, including mAP@0.5 (mean average precision with an IoU threshold above 0.5) and mAP@0.5:0.95 (mean average precision within an IoU threshold ranging from 0.5 to 0.95 with a step size of 0.05) are utilized.

B. Comparison of Results

This study seeks to ascertain if incorporating synthetic data would amplify the performance of detectors on the Heridal dataset. Furthermore, the influence of synthetic data alongside a transformer module on detector functionality is examined on

the RGBTDronePerson and VTSaR dataset. It is worth noting that due to the lack of thermal images in the Heridal dataset, verifying the effectiveness of the transformer module in this case is unfeasible.

1) *RGBTDronePerson Dataset*: The performance of VTSaRNet is compared to state-of-the-art multispectral object detection algorithms, including CFT (without mosaic and perspective data augmentation) [13], QFDet [53], as well as baseline detectors tailored for RGBT object detection: Faster R-CNN [55], Cascade R-CNN [56], RetinaNet [57], and YOLOv5 [12]. Experiments were conducted using the RGBTDronePerson dataset, with results summarized in Table II. VTSaRNet achieves a mean average precision (mAP) of 40.42% at a threshold of 0.5, and a best mAP of 30.64% at thresholds of 0.5 to 0.95. Notably, VTSaRNet outperforms the second-best competitor, QFDet, by

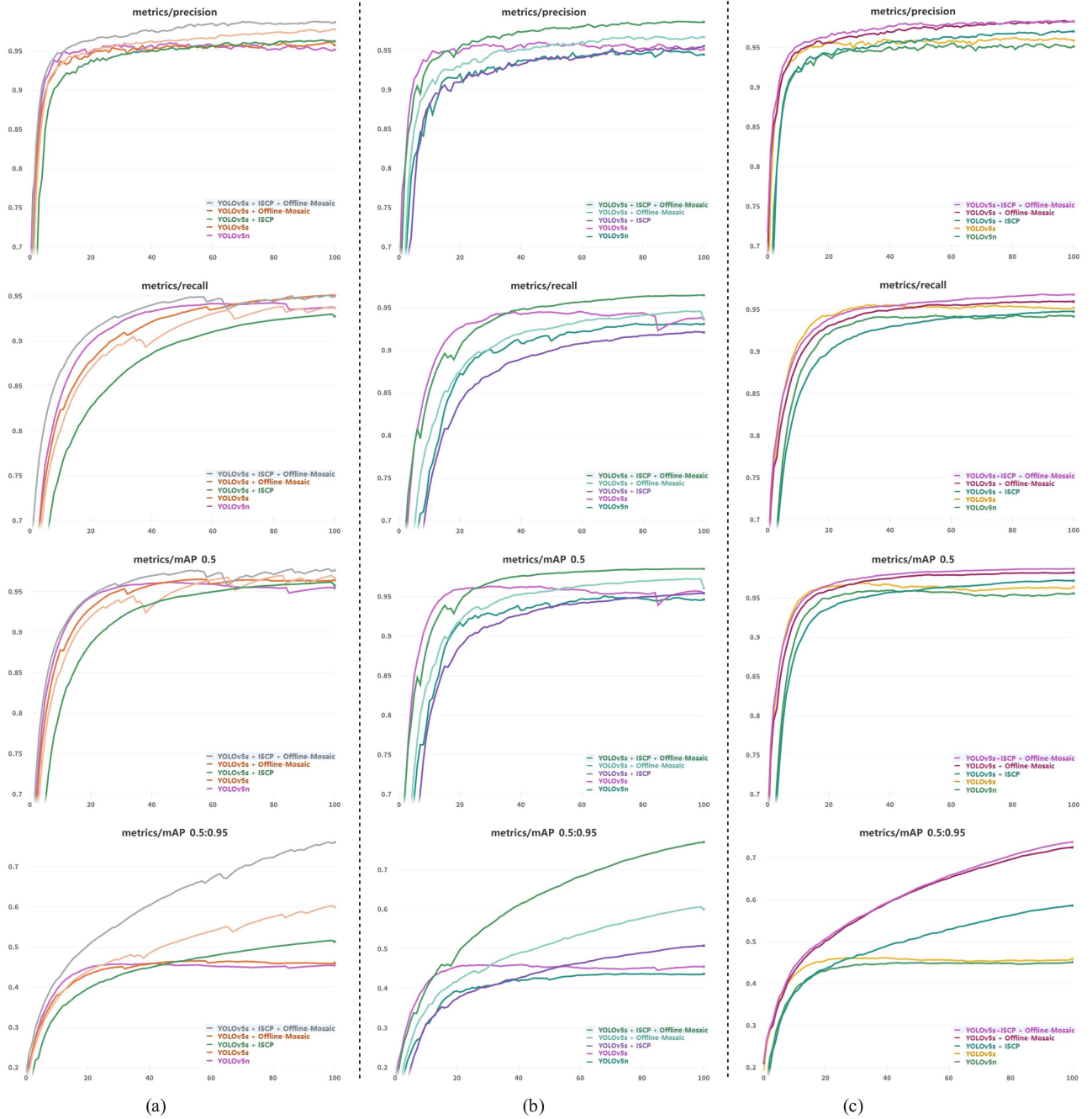


Fig. 11. Comparison of dual-modality training curves for various methods using the VTSaR dataset, based on the pretrained models labeled “s” and “n.” General metrics for person detection (precision, recall, mAP@0.5, and mAP@0.5:0.95) are represented by curves (a–c). Each curve color corresponds to a different strategy proposed by the modules. “s” and “n” indicate the depth of the pretrained model backbones. All benchmarks were tested using the YOLOv5 framework. (a) RGB. (b) IR. (c) Ours (RGB+IR).

8.17 points in mAP@0.5:0.95. In addition, VTSaRNet achieves the highest detection accuracy, with precision at 60.79% and recall at 53.66%, while maintaining a relatively small parameter count.

2) *VTSaR Dataset*: The effects of varying model depths and widths, as well as different data synthesis methods, were evaluated on self-built VTSaR dataset using YOLOv5n, YOLOv5s, and their respective variants (YOLOv5s + ISCP, YOLOv5s +

Offline-Mosaic, and YOLOv5s + ISCP + Offline-Mosaic) to assess the detection performance of the joint YOLOv5 and Transformer structure. Fig. 11 and Table III illustrate the results. First, the performance metrics of RGB and thermal modes using the YOLOv5 method along with different enhancement techniques were examined, and the metrics varied across the RGB/thermal modalities, with some improving while others declining. In general, the additional modules showed effectiveness in

TABLE III
COMPARATIVE RESULTS OF VARIOUS MODULE CONFIGURATIONS UNDER DIFFERENT PRETRAINED MODELS FOR RGB AND THERMAL MODALITIES ON THE SELF-CREATED VTSAR DATASET

Method	Modality	Batch_Size	Precision	Recall	AP ^{val} _{0.5}	AP ^{test} _{0.5}	AP ^{val} _{0.5:0.95}	AP ^{test} _{0.5:0.95}	Params /M	GFLOPs /M	Training Time /d_h_m_s
YOLOv5n	RGB	64	0.9527	0.9273	0.9488	0.9414	0.4554	0.4523	2.1	4.9	1h_19m_27s
	Thermal	64	0.9444	0.933	0.9493	0.9429	0.4399	0.4346	2.3	5.4	1h_18m_15s
	RGB+Thermal	32	0.9526	0.9374	0.9573	0.9504	0.4541	0.4529	2.1	6.8	4h_33m_2s
			0.82%	1.01%	0.85%	0.90%	1.42%	1.83%			
YOLOv5s	RGB	32	0.9546	0.9047	0.9675	0.9612	0.4679	0.4602	6.7	15.2	3h_42m_35s
	Thermal	32	0.9522	0.916	0.941	0.9376	0.4398	0.4327	6.8	15.1	3h_4m_58s
	RGB+Thermal	16	0.9574	0.9571	0.9691	0.9602	0.4665	0.4605	6.7	15.1	8h_21m_16s
			0.52%	5.24%	2.81%	2.26%	2.67%	2.78%			
YOLOv5s +ISCP	RGB	16	0.9622	0.8919	0.9223	0.9179	0.4986	0.4923	6.9	14.9	3h_47m_37s
	Thermal	16	0.9566	0.9188	0.9544	0.9488	0.5087	0.5011	6.8	15.1	3h_52m_8s
	RGB+Thermal	8	0.9704	0.9472	0.9719	0.9659	0.5874	0.5829	6.7	15.2	9h_23m_11s
			1.38%	2.84%	1.75%	1.71%	7.87%	8.18%			
YOLOv5s +Offline-Mosaic	RGB	8	0.9774	0.9491	0.9378	0.9364	0.586	0.5813	6.8	15.4	3h_33m_23s
	Thermal	8	0.9668	0.8964	0.9285	0.9211	0.5996	0.5927	6.8	15.4	3h_29m_16s
	RGB+Thermal	2	0.983	0.9591	0.9827	0.9786	0.7261	0.7189	6.7	15.2	9h_44m_20s
			1.62%	6.27%	5.42%	5.75%	12.65%	12.62%			
YOLOv5s+ISCP +Offline-Mosaic	RGB	4	0.9874	0.9616	0.9856	0.9799	0.7718	0.7684	6.8	15.1	3h_19m_26s
	Thermal	4	0.9865	0.9646	0.9853	0.9782	0.7725	0.7659	6.9	15.2	3h_49m_52s
	RGB+Thermal	2	0.983	0.9678	0.9873	0.9821	0.7398	0.7322	6.9	15.2	9h_31m_24s
			-0.35%	0.32%	0.20%	0.39%	-3.27%	-3.37%			
			2.56%	1.07%	1.82%	2.19%	27.33%	27.17%			

¹ The cells with a light green background represent the rate of increase or decrease of RGB+Thermal compared to the minimum value of the single modality (RGB or Thermal) in each method.

² The cells with a light orange background represent the rate of increase or decrease, relative to the baseline (without any improvement strategies), of each method under the RGB+Thermal modality using the yolov5s pretrained model.

enhancing the detection indicators. Second, when performing dual-mode fusion detection, a noticeable improvement was observed compared to single-mode detection. Specifically, AP@0.5:0.95 demonstrated a maximum effective increase of 12.62%. Moreover, the fusion metrics of all detectors in YOLOv5s + ISCP + Offline-Mosaic reached Precision = 0.983, Recall = 0.9678, AP@0.5 = 0.9873, and AP@0.5:0.95 = 0.7398, respectively. These improvements were 2.56%, 1.07%, 1.82%, and 27.33% higher than those of YOLOv5s.

To analyze the attention of the object detector on detail features, visualization modules were incorporated in the 17th, 20th, and 23rd convolutional layers of the inference stage. This allowed us to observe changes in the intermediate feature map. Fig. 12 illustrates the thermal diagram results for both the RGB and thermal modality on the output feature map of these three layers. The visualizations reveal that as the network progresses, the extraction of target features becomes more accurate. For instance, in conv_23, the heat map highlights areas where the object is present or where there is high heat. Compared to the RGB mode, the thermal mode produces a better characteristic heat map in conv_23 but shows room for improvement in conv_17 and conv_20. Following the fusion of the dual mode at the feature layer, as depicted on the right side of Fig. 12, the heat map demonstrates complementary characteristics from the

dual-source images. This results in a wider extracted target area, providing further evidence of the superior feature extraction ability of this structure.

C. Discussion Study

The scale and location of the synthetic data instances have a significant impact on the IoU. Furthermore, the performance of the detector varies depending on different fusion strategies. A series of discussion experiments were conducted based on these two points.

1) *Effect of ISCP With Synthesis Instance Scales*: To effectively evaluate the impact of synthetic data on detection algorithm performance, this section conducts experiments using the Heridal dataset to test three algorithms: YOLOv5, YOLOv5 + Aug, and Ours + Aug. The “Aug” refers to synthetic data augmentation methods employed, such as ISCP and offline-based Mosaic. The results, shown in Table IV and Fig. 13, indicate an overall improvement in detection performance with the enhanced model, particularly in AP@0.5:0.95 which improves by 6.77%. The approach(Ours + Aug) achieves metrics of precision = 0.8914, recall = 0.7389, AP@0.5 = 0.8115, and AP@0.5:0.95 = 0.5436. Nevertheless, in order to achieve a compromise between precision and efficiency, further studies concentrate on the “s” arrangement. It is important to note that, for some

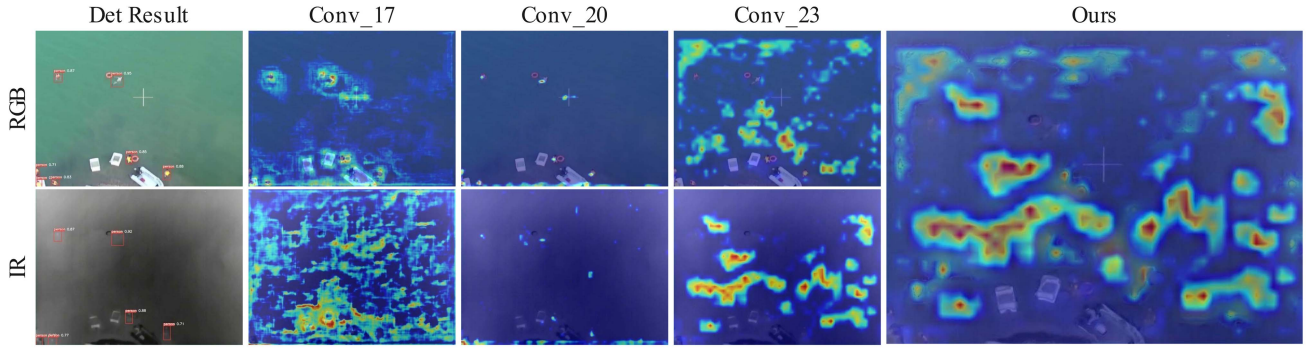


Fig. 12. Two additional instances of feature map visualization in VTSaNet, using the VTSaR dataset. The first column showcases the GT fused with detection outcomes yielded by VTSaNet. The feature maps depicted in the second, third, and fourth columns are derived from processing the RGB and thermal inputs with the Conv_17, Conv_20, and Conv_23 layers respectively. Lastly, the final column visually illustrates the feature map following the BFT module. It is pivotal to note that in every occurrence, the color branch is represented in the top example, while the thermal branch is denoted in the lowermost example. A zoom function can be employed to scrutinize and understand intricate aspects in greater depth.

TABLE IV
COMPARATIVE RESULTS OF YOLOV5 AND VTSARNET WITH DATA AUGMENTATION UNDER DIFFERENT PRETRAINED MODELS
ON THE PUBLIC-AVAILABLE HERIDAL DATASET

Method	Pretrained Model	Batch_Size	Precision	Recall	AP ^{val} _{0.5}	AP ^{test} _{0.5}	AP ^{val} _{0.5:0.95}	AP ^{test} _{0.5:0.95}	Params /M	GFLOPs /M	Training Time /d_h_m_s
YOLOv5	x	4	0.8926	0.6914	0.7962	0.7854	0.5311	0.526	85.4	204.9	1d_2h_20m_41s
	l	8	0.8592	0.7002	0.7795	0.7659	0.492	0.4869	44.9	108.6	11h_45m_27s
	m	16	0.8679	0.6439	0.752	0.7428	0.493	0.4826	19.8	47.4	9h_34m_49s
	s	32	0.8386	0.6795	0.7176	0.7002	0.4109	0.4005	6.7	15.2	5h_43m_37s
YOLOv5+Aug	x	4	0.8706 -2.20%	0.7329 4.15%	0.8079 1.17%	0.7963 1.09%	0.5405 0.94%	0.5349 0.89%	85.4	204.9	1d_7h_54m_21s
	l	8	0.8676 0.84%	0.7003 0.01%	0.7841 0.46%	0.7752 0.93%	0.5115 1.95%	0.5048 1.79%	44.9	108.6	11h_46m_46s
	m	16	0.8867 1.88%	0.6736 2.97%	0.7619 0.99%	0.7541 1.13%	0.503 1.00%	0.4812 -0.14%	19.8	47.4	9h_26m_3s
	s	32	0.8139 -2.47%	0.6884 0.89%	0.7233 0.57%	0.7168 1.66%	0.4419 3.10%	0.4351 3.46%	6.7	15.2	5h_33m_5s
VTSaNet+Aug	x	4	0.8914 -0.12%	0.7389 4.75%	0.8115 1.53%	0.8007 1.53%	0.5436 1.25%	0.5396 1.36%	85.4	204.9	1d_14h_11m_39s
	l	8	0.8741 1.49%	0.7062 0.60%	0.7852 0.57%	0.7796 1.37%	0.5182 2.62%	0.5077 2.08%	44.9	108.6	11h_34m_41s
	m	16	0.8872 1.93%	0.7003 5.64%	0.7703 1.83%	0.7653 2.25%	0.5142 2.12%	0.5065 2.39%	19.8	47.4	9h_17m_56s
	s	32	0.8693 3.07%	0.6914 1.19%	0.7343 1.67%	0.7285 2.83%	0.4713 6.04%	0.4682 6.77%	6.7	15.2	5h_19m_8s

¹ The cells with a light green background represent the rate of increase or decrease, relative to the baseline, of each method with different pretrained models [s, m, l, x]. The comparison is based on the corresponding pretrained model for each method.

indicators, there is a decline after applying the enhancement method, mainly in the precision indicator. This may be attributed to inaccuracies in the person instance segmentation model, resulting in incomplete segmentation or the inclusion of redundant background, thereby increasing the detection difficulty with the ISCP module. Since “Ours+Aug” method uses dual-source input, the Heridal dataset contains only RGB images. To accommodate dual-source input, the conversion of certain RGB modes into grayscale images supports the dual-source structure.

The Heridal dataset’s patch library generates more synthetic person samples by the ISCP module, encompassing various

scales and object numbers. Table V compares the influence of instance sizes and numbers on detector performance across different models to generate effective positive samples, the cells with a light red background indicate the cases where none of our proposed data augmentation methods were applied. On the other hand, the cells with a light green background represent the comparisons of different implanting strategies for instance sizes of 32×32 . Analysis of Table V demonstrates strong detection performance metrics when maintaining instances near a scale of 32×32 and having approximately 4–7 instances. Notable consistency in performance across distinct datasets—Heridal and VTSaR—has been observed. In addition, the utilization

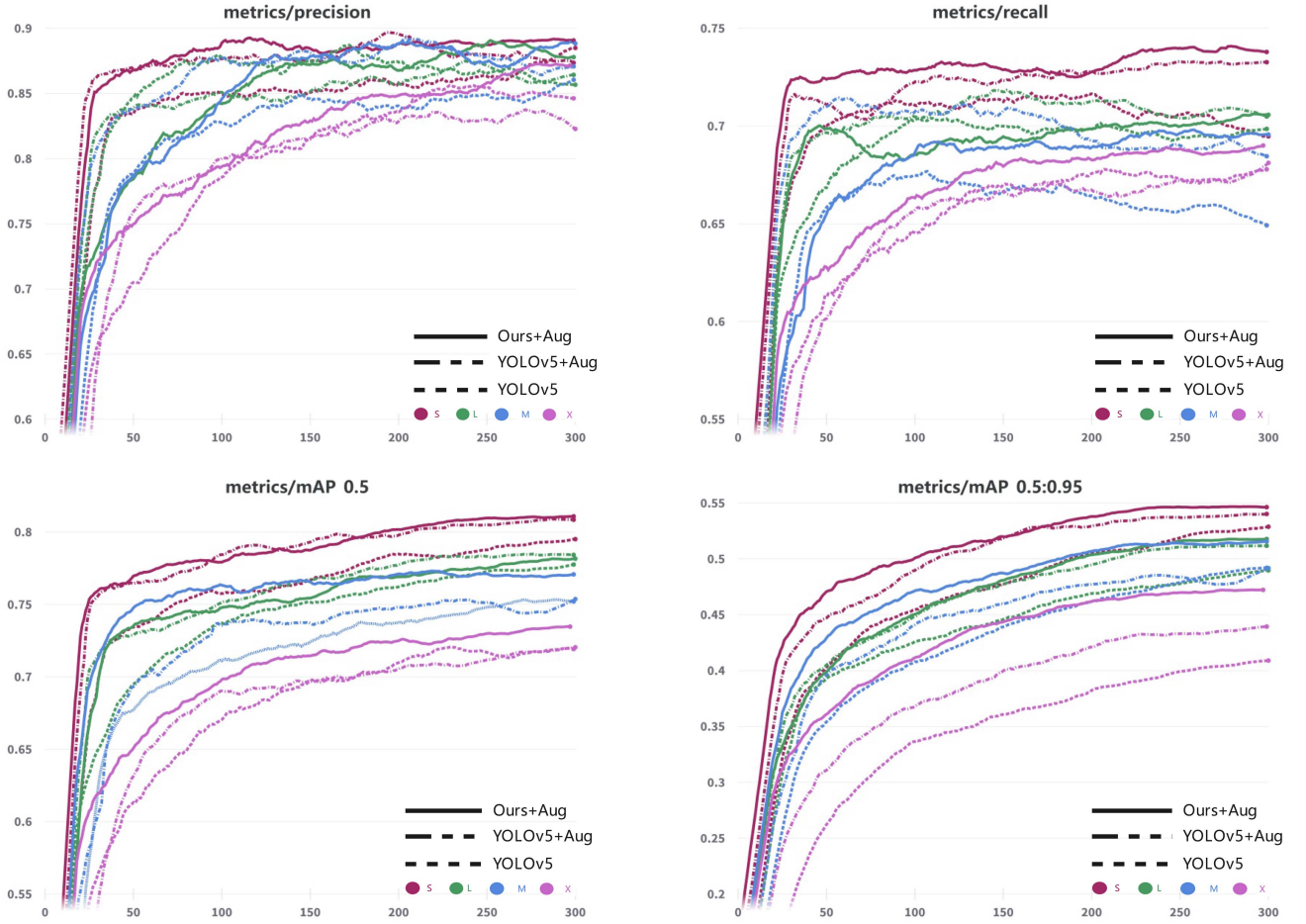


Fig. 13. Training curves on the Heridal dataset were generated using models of various depths, including baseline models and our proposed methods. The curves were evaluated based on metrics such as precision, recall, mAP@0.5, and mAP@0.5:0.95.

TABLE V
EFFECTS OF ISCP AND MOSAIC MODULES ON TRAINING METRICS OF MAP

Method	Dataset	ISCP			Mosaic			mAP
		Segment Model	Instance Size	Num (random)	Repeat	Online	Offline	
Baseline	Heridal	x	x	x	x	√	x	0.4109
		s-seg	32x32	[4, 7]	x	√	x	0.3842
		s-seg	32x32	[4, 7]	x	√	√	0.4419
		s-seg	32x32	[4, 7]	√	√	√	0.4253
		s-seg	32x32	[8, 12]	x	√	√	0.4167
		s-seg	32x32	[13, 20]	x	√	√	0.3658
		s-seg	16x16	[4, 7]	x	√	√	0.3726
		s-seg	64x64	[4, 7]	x	√	√	0.3841
		s-seg	81x81	[4, 7]	x	√	√	0.3968
		s-seg	128x128	[4, 7]	x	√	√	0.3901
VTSaRNet (Ours)	VTSaR	x	x	x	x	√	x	0.4679
		s-seg	32x32	[4, 7]	x	√	x	0.4986
		s-seg	32x32	[4, 7]	x	√	√	0.7718
		s-seg	32x32	[4, 7]	√	√	√	0.7546
		s-seg	32x32	[8, 12]	x	√	√	0.7239
		s-seg	32x32	[13, 20]	x	√	√	0.7451
		s-seg	16x16	[4, 7]	x	√	√	0.7368
		s-seg	64x64	[4, 7]	x	√	√	0.7282
		s-seg	81x81	[4, 7]	x	√	√	0.7356
		s-seg	128x128	[4, 7]	x	√	√	0.7004

of both offline and online Mosaic enhancement facilitates the incorporation of a greater quantity of positive samples within the training process, as corroborated by the comparison results in Table V. Furthermore, the two data synthesis methods

used serve to boost the training performance of the detector without resulting in a considerable increase in processing time.

2) *Effect of BFT With Different Fusion Strategies*: BFT are compared with different resolutions using three fusion strategies, as presented in Table VI. The initial models were developed, which had single-modality detectors for RGB and thermal data. As depicted in Fig. 3, Input-Fusion I encompasses the process of pixel-level summing and channel-level concatenation. Mid-Fusion II consists of two-stream YOLOv5s architectures, similar to the ones shown in Fig. 3, with the BFT module replaced by summation and concatenation. Lastly, Last Fusion III performs a weighted fusion of the results and applies nonmaximum suppression (NMS) to filter the final bounding box. Table VI presents the metrics computed during the training stage, with the best value for each metric highlighted in bold. The evaluation metrics, namely, precision, recall, mAP@0.5, and mAP@0.5:0.95, demonstrated remarkable performance on the VTSaR dataset when utilizing the YOLOv5s pretrained model. Specifically, the employment of the Mid-Fusion II method and high resolution yielded impressive values of 98.81%, 96.78%, 98.75%, and 73.98% for the aforementioned metrics, respectively. This demonstrates that superior performance metrics can be attained

TABLE VI
EFFECT OF BFT WITH VARIOUS FUSION STRATEGIES ON TRAINING METRICS FOR THE VTSAR DATASET

Model	Resolution	Fusion Strategies			Metrics		
		ef	mf	lf	Precision	Recall	mAP@0.5 mAP@0.5:0.95
YOLOv5s	1536×1536	✓			0.952	0.9414	0.9529
	1536×1536		✓		0.983	0.9678	0.9873
	1536×1536			✓	0.9374	0.9187	0.9685
	1024×1024	✓			0.9522	0.9155	0.9476
	1024×1024		✓		0.9881	0.9488	0.9875
	1024×1024			✓	0.9489	0.8856	0.9521
	640×640	✓			0.9168	0.8769	0.932
	640×640		✓		0.9459	0.9132	0.9453
	640×640			✓	0.8747	0.8697	0.9106

TABLE VII
COMPARISON OF THE DETECTION METRICS OF VTSARNET WITH STATE-OF-THE-ART (SOTA) METHODS

DataSets	OA&Our Methods	InputSize /pix	BackBone	PreTrainedModel	Precision /%	Recall /%	AP ^{val} @0.5 /%	AP ^{val} @0.5:0.95 /%	InferTime ^{test} /ms (RTX4090)
Heridal	SSD [58]	1333x1000	VGG16	ssd512_coco	4.33	94.36	-	-	22.6
	Faster RCNN [55]	1333x1000	ResNet101+FPN	faster_rcnn_r101_fpn_lx_coco	58.1	85	-	-	206.8
	Božić-Štulić et al. [1]	1333x1000	RPM+SOD	faster_rcnn_r101_fpn_lx_coco	34.8	88.9	-	-	-
	RPN [59]	4000x3000	ResNet101+FPN	rpn_x101_32x4d_fpn_lx_coco	41.54	95.54	-	-	-
	RFCCD [60]	4000x3000	ResNet101+FPN	rpn_x101_32x4d_fpn_lx_coco	68.89	94.65	-	-	-
	Zhang et al. [61]	1536x1536	DarkNet-53+ Focus	yolov5s	80.69	67.84	71.29	40.47	9.3
	Zhang et al.	1536x1536	DarkNet-53+ Focus	yolov5x6	82.06	68.46	72.51	43.69	79.4
	YOLOv5 [12]	1536x1536	DarkNet-53+ Focus	yolov5s	83.86	67.95	71.76	41.09	9.6
	YOLOv5+Aug (ours)	1536x1536	DarkNet-53+ Focus	yolov5s	81.39	68.84	72.33	44.19	9.8
	VTSaRNet (ours) /rgb modality	1536x1536	DarkNet-53+ Focus	yolov5s	86.93	69.14	73.43	47.13	9.5
VTSaR	SSDMNV2 [62]	512x512	MobileNet v2	ssd512_mobilenetv2_600e_coco	57.29	47.62	56.26	15.4	12.4
	RetinaNet [57]	640x512	ResNet101+FPN	retinanet_r50_fpn_lx_coco	91.88	91.29	88.42	37.26	147.2
	Faster RCNN [55]	640x512	ResNet50+FPN	faster_rcnn_r101_fpn_lx_coco	93.21	94.88	96.87	44.82	211.8
	Faster RCNN [55]	1536x1536	ResNet50+PANet+FPN	faster_rcnn_r101_panet_fpn_lx_coco	95.46	96.53	95.41	48.67	237.5
	FCOS [63]	1536x1536	ResNet50	fcos_r50_caffe_fpn_gn_head_lx_coco	92.13	91.52	96.59	50.04	44.9
	Swin-T [64]	1024x1024	ResNet50	retinanet_swin-t-p4-w7_fpn_lxcoco	95.86	94.67	97.52	65.12	104.8
	YOLOv4 [47]	1600x1600	CSPDarknet53-PANet-SPP	yolov4-csp-panet-spp-x-swish	92.51	92.14	94.67	58.74	35.2
	YOLOv5/rgb modality	1536x1536	DarkNet-53+ Focus	yolov5n	95.27	92.73	94.88	45.54	3.9
	YOLOv5 [12] /rgb modality	1536x1536	DarkNet-53+ Focus	yolov5s	95.46	90.47	96.75	46.79	9.4
	YOLOv5+ISCP (ours) /rgb modality	1536x1536	DarkNet-53+ Focus	yolov5s	96.22	89.19	92.23	49.86	9.6
	YOLOv5+Mosaic (ours) /rgb modality	1536x1536	DarkNet-53+ Focus	yolov5s	97.74	94.91	93.78	58.6	9.6
	VTSaRNet (ours) /rgb modality	1536x1536	DarkNet-53+ Focus	yolov5s	98.74	96.16	98.56	77.18	9.7
	VTSaRNet (ours) /dual modality	1536x1536	DarkNet-53+ Focus	yolov5s	98.3	96.78	98.73	73.98	16.4

through the use of the Mid-Fusion II strategy and higher resolution.

D. Comparisons With the State-of-the-Art

To assess the robustness and effectiveness of VTSaRNet on the Heridal and VTSaR datasets, separately from the state-of-the-art (SOTA) algorithm: SSD [58], Faster RCNN [55], Božić-Štulić et al. [1], RPN [59], RFCCD [60], Zhang et al. [61], YOLOv5 [12], SSDMNv2 [62], RetinaNet [57], FCOS [63], Swin-T [64], and YOLOv4 [47] obtained on the original Public test results were obtained from the original literature work, while training metrics on the self-built dataset were evaluated using methods employed in this study. In addition, other SOTA test results were obtained using the MMDetection toolbox. Table VII illustrates the significant improvement achieved by VTSaRNet in detecting both the Heridal and VTSaR datasets, thereby establishing a foundation for the commercial application of SaR tasks.

E. Visualization

Figs. 14 and 15 display visualization results for various baseline detectors and VTSaRNet across five scenarios, drawn from the publicly available Heridal dataset and the self-constructed VTSaR dataset. The detection outcomes reflect a marked performance improvement relative to the currently acknowledged baseline detectors. Specifically, VTSaRNet exhibits notable FN improvement. Baseline detectors commonly suffer from low FN when detecting persons due to the absence of feature regression information. Moreover, when the anchor-based detector is equipped with the YOLO series, it can effectively learn target information from dual-source positive small samples. Moreover, it is worth noting that a considerable quantity of false positives can be identified, suggesting that SSD has a deficiency in reliably distinguishing correct predictions from a large pool of detection possibilities. Remarkably, VTSaRNet correctly handles FP detection, signifying the higher quality of assigned positive/negative samples.

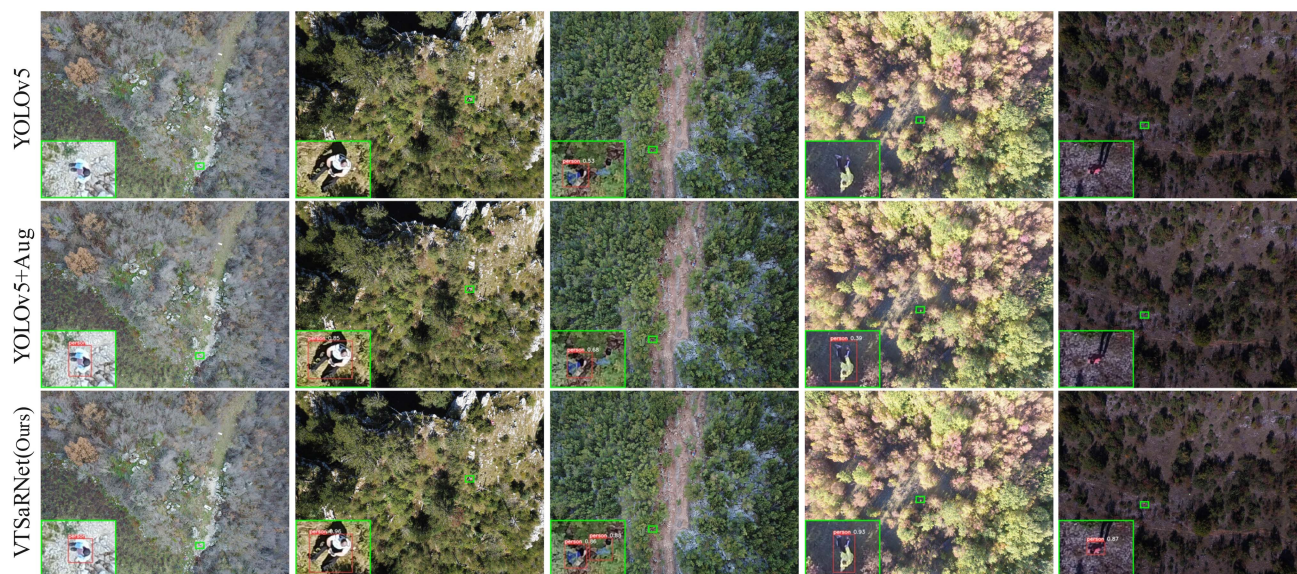


Fig. 14. Detection results of advanced detectors (first two rows) and VTSaRNet (the third row) on the publicly available Heridal dataset are visualized. Partial enlargements of the test results from various methods are presented for demonstration purposes.

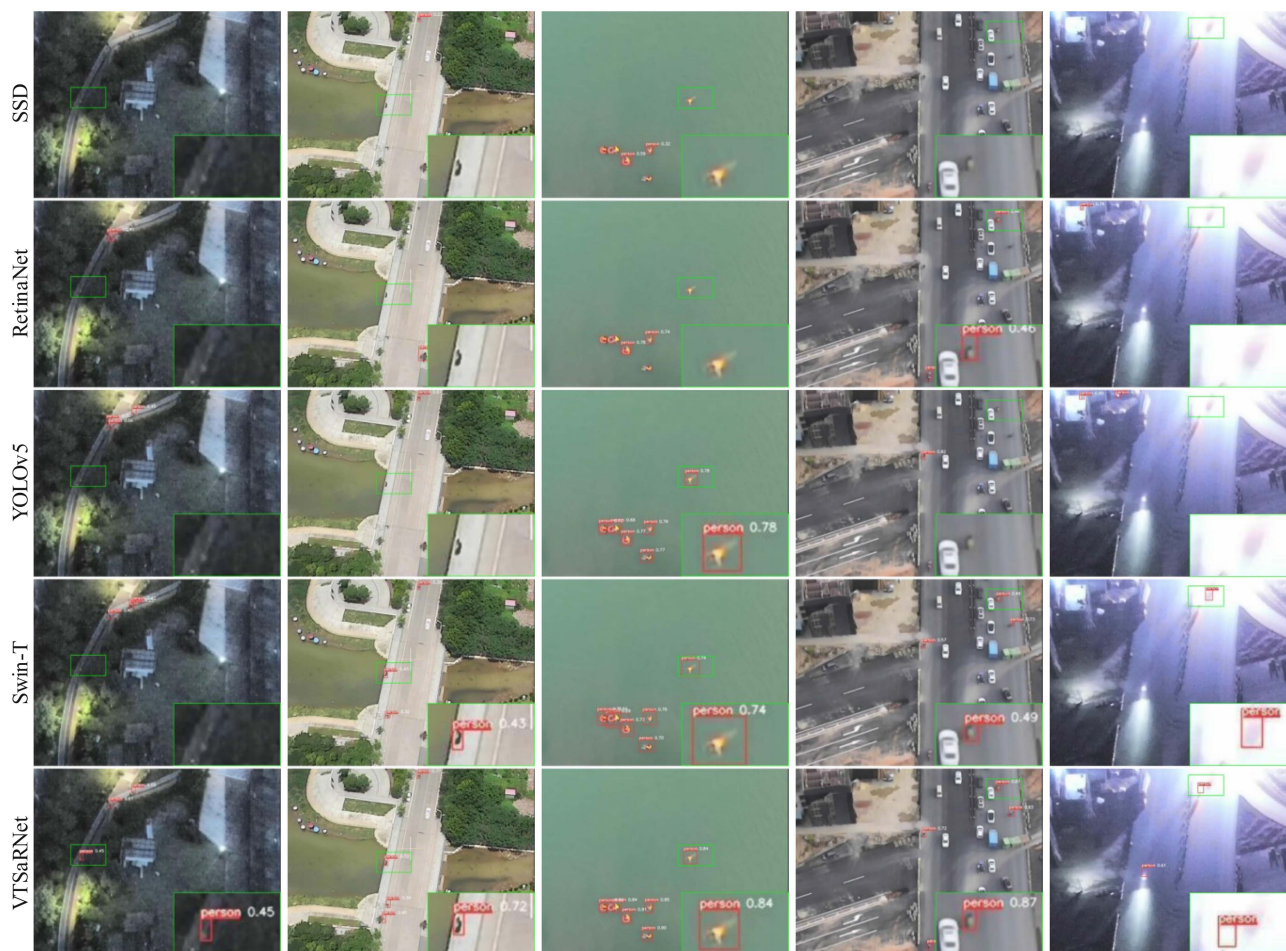


Fig. 15. Visualization detection results of advanced detectors (first four rows) and VTSaRNet (the fifth row) on the self-created VTSaR dataset. Partial enlargements of the test results from the various methods are included for demonstration purposes.

V. CONCLUSION

This article explores two key challenges in detecting persons in aerial images with a broad field-of-view: the samples exhibit a faint sparsity attribute alongside an inadequate representation of single-modal features. To address these limitations, we emphasize the efficient detection of persons via block-based sparse learning, coupled with the progressive fusion detection of personnel features implemented by a singular encoder-decoder detector. Experimental results on the public Heridal dataset demonstrate significant performance improvements, reaching a mAP@0.5 of 81.15%, which is a 1.53% improvement over YOLOv5 (Since Heridal lacks a thermal modality, the transformer module could not be tested. The improvement solely comes from optimizing data augmentation). On the self-built VTSaR dataset, the detection performance of multimodal approaches significantly outperforms single-modal, particularly with a mAP@0.5:0.95 of 73.98% achieved under multimodal conditions, showing a remarkable 27.33% improvement over YOLOv5s. However, The shape of small objects (e.g., persons) may vary with rapid changes in angles, poses, and degrees of occlusion from the perspective of broad and high SaR scenarios. This leads to diverse appearances of the persons, increasing the difficulty of accurate localization and classification. In the future, the proposed method for detecting persons in aerial imagery will be extended to other aerial reconnaissance tasks, such as cross-modal object detection, anomaly event detection, fine-grained detection, and object tracking. This research has important implications for enhancing maneuverable search, object detection, and national defense in unmanned aerial SaR systems for reconnaissance, counter-terrorism, surveillance, and public security.

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